

Week - 4

2520030030

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1. CREATE DATABASE

```
1 • DROP DATABASE IF EXISTS bookflow_db;
2 • CREATE DATABASE bookflow_db;
3 • USE bookflow_db;
```

Output				
Action Output				
#	Time	Action	Message	
✓ 1	19:37:24	DROP DATABASE IF EXISTS bookflow_db	5 row(s) affected	
✓ 2	19:37:26	CREATE DATABASE bookflow_db	1 row(s) affected	
✓ 3	19:37:27	USE bookflow_db	0 row(s) affected	

2. CREATE BOOKS TABLE

```
4 • CREATE TABLE Books (
5     book_id INT AUTO_INCREMENT PRIMARY KEY,
6     title VARCHAR(200) NOT NULL,
7     author VARCHAR(150) NOT NULL,
8     genre VARCHAR(100),
9     description TEXT,
10    published_year INT
11 );
12
```

Output				
Action Output				
#	Time	Action	Message	
✓ 1	19:37:24	DROP DATABASE IF EXISTS bookflow_db	5 row(s) affected	
✓ 2	19:37:26	CREATE DATABASE bookflow_db	1 row(s) affected	
✓ 3	19:37:27	USE bookflow_db	0 row(s) affected	
✓ 4	19:38:29	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NUL...	0 row(s) affected	

3. INSERT BOOK DATA

```
12 • INSERT INTO Books(title, author, genre, description, published_year) VALUES
13 (
14     'The Martian',
15     'Rohit Sai',
16     'Science Fiction',
17     'An astronaut stranded on Mars must survive using science and engineering.',
18     2011
19 ),
20 (
21     'Dune',
22     'Frank Herbert',
23     'Science Fiction',
24     'A young nobleman becomes involved in the politics and destiny of a desert planet.',
25     1965
26 ),
27 ;
```

Output				
Action Output				
#	Time	Action	Message	
✓ 1	19:37:24	DROP DATABASE IF EXISTS bookflow_db	5 row(s) affected	
✓ 2	19:37:26	CREATE DATABASE bookflow_db	1 row(s) affected	
✓ 3	19:37:27	USE bookflow_db	0 row(s) affected	
✓ 4	19:38:29	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NUL...	0 row(s) affected	
✓ 5	19:40:07	INSERT INTO Books(title, author, genre, description, published_year) VALUES ('The Martian', 'Rohit Sai', '...'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	

4. DISPLAY BOOKS

```
83 • SELECT
84     book_id,
85     title,
86     author,
87     genre,
88     published_year
89 FROM Books
90 ORDER BY book_id;
```

Result Grid

	book_id	title	author	genre	published_year
▶	1	The Martian	Rohit Sai	Science Fiction	2011
	2	Dune	Frank Herbert	Science Fiction	1965
	3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
	4	Pride and Prejudice	Jane Austen	Romance	1813
	5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
	6	Foundation	Isaac Asimov	Science Fiction	1951
	7	The Hobbit	J.R. Tolkien	Fantasy	1937
	8	Enders Game	Orson Scott Card	Science Fiction	1985
	9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997
	10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870
•	NULL	NULL	NULL	NULL	NULL

Books 1 x

Output

Action Output

#	Time	Action	Message
✓	2 19:37:26	CREATE DATABASE bookflow_db	1 row(s) affected
✓	3 19:37:27	USE bookflow_db	0 row(s) affected
✓	4 19:38:29	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
✓	5 19:40:07	INSERT INTO Books(title, author, genre, description, published_year) VALUES ('The Martian', 'Rohit Sai', ...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓	6 19:41:59	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, ...	10 row(s) returned

5. ADD EMBEDDING COLUMN

```
91 • ALTER TABLE Books
92     ADD COLUMN embedding JSON;
93
94
95
```

Output

Action Output

#	Time	Action	Message
▶	3 19:37:27	USE bookflow_db	0 row(s) affected
▶	4 19:38:29	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
▶	5 19:40:07	INSERT INTO Books(title, author, genre, description, published_year) VALUES ('The Martian', 'Rohit Sai', ...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
▶	6 19:41:59	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, ...	10 row(s) returned
▶	7 20:05:31	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

6. INSERT MOCK VECTORS

```
105 WHERE book_id = 3;
106 • UPDATE Books
107 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
108 WHERE book_id = 4;
109 • UPDATE Books
110 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
111 WHERE book_id = 5;
112 • UPDATE Books
113 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
114 WHERE book_id = 6;
115 • UPDATE Books
116 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
117 WHERE book_id = 7;
118 • UPDATE Books
119 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
120 WHERE book_id = 8;
121 • UPDATE Books
122 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
123 WHERE book_id = 9;
124 • UPDATE Books
125 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
126 WHERE book_id = 10;
```

Output

Action Output

#	Time	Action	Message
✓ 13	20:07:38	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 14	20:07:40	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 15	20:07:42	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 16	20:07:44	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 17	20:07:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

7. VERIFY ALL VECTORS

```
128 • SELECT
129     book_id,
130     title,
131     embedding
132 FROM Books
133 ORDER BY book_id;
134
135
```

Result Grid

	book_id	title	embedding
▶	1	The Martian	[0.90, 0.80, 0.20]
	2	Dune	[0.85, 0.75, 0.25]
	3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
	4	Pride and Prejudice	[0.10, 0.20, 0.90]
	5	The Great Gatsby	[0.15, 0.30, 0.80]
	6	Foundation	[0.95, 0.65, 0.20]
	7	The Hobbit	[0.30, 0.75, 0.90]
	8	Enders Game	[0.88, 0.85, 0.15]
	9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
	10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]
*	NULL	NULL	NULL

Books 2 x

Output

Action Output

#	Time	Action	Message
✓ 14	20:07:40	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 15	20:07:42	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 16	20:07:44	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 17	20:07:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 18	20:08:55	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned

8. L2 DISTANCE -- Compare every book with Book 1 vector: -- [0.90, 0.80, 0.20]

```
134 • SELECT
135     book_id,
136     title,
137     author,
138     genre,
139     SQRT(
140         POW(
141             CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[0]')) AS DECIMAL(10,5))
142             - 0.90,
143             2
144         )
145         +
146         POW(
147             CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[1]')) AS DECIMAL(10,5))
148             - 0.80,
149             2
150         )
151         +
152         POW(
153             CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[2]')) AS DECIMAL(10,5))
154             - 0.20,
155             2
156         )
157     ) AS l2_distance
158 FROM Books
159 WHERE embedding IS NOT NULL
160 ORDER BY l2_distance ASC
161 LIMIT 3;
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Contents: | Fetch rows:

	book_id	title	author	genre	l2_distance
▶	1	The Martian	Rohit Sai	Science Fiction	0
	8	Enders Game	Orson Scott Card	Science Fiction	0.07348469228349536
	2	Dune	Frank Herbert	Science Fiction	0.08660254037844388

Result 3 x

Output

Action Output

#	Time	Action	Message
✓ 15	20:07:42	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 16	20:07:44	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 17	20:07:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 18	20:08:55	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned
✓ 19	20:10:07	SELECT book_id, title, author, genre, SQRT(POW(CAST(JSON_UNQUOTE(JSON_...	3 row(s) returned

9. COSINE SIMILARITY -- Compare every book with Book 1 vector: -- [0.90, 0.80, 0.20]

```

162 • SELECT
163     book_id,
164     title,
165     author,
166     genre,
167     (
168         (
169             CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[0]')) AS DECIMAL(10,5)) * 0.90
170         +
171             CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[1]')) AS DECIMAL(10,5)) * 0.80
172         +
173             CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[2]')) AS DECIMAL(10,5)) * 0.20
174         )
175         /
176         (
177             Sqrt(
178                 POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[0]')) AS DECIMAL(10,5)), 2)
179             +
180                 POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[1]')) AS DECIMAL(10,5)), 2)
181             +
182                 POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[2]')) AS DECIMAL(10,5)), 2)
183             )
184         )

```

```

184         *
185         Sqrt(
186             POW(0.90, 2)
187         +
188             POW(0.80, 2)
189         +
190             POW(0.20, 2)
191         )
192     )
193 ) AS cosine_similarity
194 FROM Books
195 WHERE embedding IS NOT NULL
196 ORDER BY cosine_similarity DESC
197 LIMIT 3;

```

book_id	title	author	genre	cosine_similarity
1	The Martian	Rohit Sai	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986169165587373
8	Enders Game	Orson Scott Card	Science Fiction	0.9982532606477712

#	Time	Action	Message
✓ 16	20:07:44	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 17	20:07:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 18	20:08:55	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned
✓ 19	20:10:07	SELECT book_id, title, author, genre, Sqrt(POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '\$[0]')) AS DECIMAL(10,5)), 2) + POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '\$[1]')) AS DECIMAL(10,5)), 2) + POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '\$[2]')) AS DECIMAL(10,5)), 2)) FROM Books	3 row(s) returned
✓ 20	20:12:58	SELECT book_id, title, author, genre, ((CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '\$[0]')) AS DECIMAL(10,5)) * 0.90 + CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '\$[1]')) AS DECIMAL(10,5)) * 0.80 + CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '\$[2]')) AS DECIMAL(10,5)) * 0.20) / (Sqrt(POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '\$[0]')) AS DECIMAL(10,5)), 2) + POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '\$[1]')) AS DECIMAL(10,5)), 2) + POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '\$[2]')) AS DECIMAL(10,5)), 2))) AS cosine_similarity FROM Books WHERE embedding IS NOT NULL ORDER BY cosine_similarity DESC LIMIT 3;	3 row(s) returned